MVP DEFINITION

Landscape Health Monitoring at the Humber Arboretum

Field App + Staff Dashboard

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1. What Is the MVP and Why

An MVP (Minimum Viable Product) is the simplest version of a product that still proves the core idea works. For this project we are not trying to build everything in four weeks. We are building the smallest version of the platform that proves one thing: a visitor can log an observation in the field, and that data ends up in a system that staff can act on.

We have two MVPs that work together as one system:

- Field App MVP: a mobile web app for visitors to log in, explore the map, view tree info, and submit observations
- Staff Dashboard MVP: a web dashboard for sustainability staff to see the landscape inventory and review visitor submissions

The core loop

Visitor logs in → finds a tree on the map → submits a photo observation → staff sees it on the dashboard → staff verifies it → it becomes part of the permanent landscape record.

2. What Data the Platform Works With

The platform works with two types of data that appear on the same map at the same time.

Existing landscape inventory

This is the pre-existing dataset from the TRCA 2021 Terrestrial Biological Inventory. It covers the full 104 hectares of the Humber Arboretum and includes trees, shrubs, water bodies, trails, buildings, and other landscape elements. It is imported into Supabase at the start and shown on the staff map as the base layer.

Each element has a type (Tree, Shrub, Water Body, Trail, Building etc.), a classification (Softscape, Hardscape, or Hybrid), GPS coordinates, and attributes specific to its type — for trees: species name, native/invasive status, height, and trunk diameter.

Visitor observations

These come in through the Field App in real time. When a visitor submits an observation it is stored in Supabase with a status of pending. Staff review it on the dashboard and either verify it — making it part of the landscape inventory or flag it for follow-up.

Each observation contains a photo, species name, element type, GPS coordinates captured automatically, the visitor's user ID, a timestamp, optional notes, and a status of pending or verified.

> **Why this matters**
> Visitor observations start as unverified community data. Once a staff member reviews and verifies them, they become part of the official landscape inventory. This is how the platform turns citizen science into credible institutional data without any manual transfer.

3. Field App 6 Screens

The Field App is a mobile web app built in Next.js. It is designed to be used on a smartphone by visitors walking around the arboretum. It has six screens.

01
All users

Login

The entry point. Required for security and to link observations to a known user in Supabase.

- Email and password login via Supabase Auth
- No sign up in the MVP — accounts are created by staff or admin
- Redirects to the map on successful login

02
Visitors + on-site staff

Map

The home screen after login. Shows the real Humber Arboretum with observation pins from the database.

- Real OSM map via Leaflet.js, centred on the arboretum
- Green pins for logged observations pulled from Supabase
- Tap a pin to open the tree detail sheet
- Add observation button to start the submission flow

03
Visitors

Tree Detail

A bottom sheet that slides up when a visitor taps a pin. Shows what is known about the tree at that location.

- Common name and scientific name
- Native or introduced status
- Fun fact about the species
- CO_2 absorbed per year
- Wildlife it supports
- Health status and score
- Button to log an observation at this location

04
Visitors

Camera

The photo capture screen. Visitors point at what they see and take a photo or upload from their gallery.

- Live camera view via the device browser API
- Shutter button to capture
- Upload from gallery option
- AI species suggestion shown as an overlay label

05
Visitors

Add Entry Form

The data entry screen. Visitors confirm details about their observation before submitting.

- Photo preview from the camera screen
- Name field with AI suggestion pre-filled and editable
- Landscape element dropdown — Tree, Plant, Bird, Other
- Location — GPS coordinates captured automatically
- Optional notes field
- Discard and Save buttons

06
Visitors

Confirmation

Shown after a successful submission. Lets the visitor know their observation was received.

- Success message with observation summary
- Species name, zone, and AI match confidence
- Status: visible on map and pending staff review
- Back to map and log another observation buttons

4. Staff Dashboard — 4 Screens

The Staff Dashboard is a web application built in Next.js, designed for the Humber Office of Sustainability on desktop or tablet. It has four screens.

01
Sustainability staff

Login

The entry point. Staff log in with institutional credentials. Staff accounts have read and write access to all observations.

- Email and password login via Supabase Auth
- Staff role verified on login
- Redirects to the dashboard map on success

02
Sustainability staff

Dashboard Map

The main screen. An interactive OSM map showing the full arboretum with two data layers at the same time.

- OpenStreetMap base layer via Leaflet.js
- Layer 1 — existing landscape inventory, colour-coded by type
- Layer 2 — visitor observations as a distinct pin style
- Layer toggles to show or hide each layer independently
- Pending review count shown in the header
- Click any element or pin to open the detail panel

03
Sustainability staff

Element Detail Panel

Opens when staff click any element on the map. Works for both inventory elements and visitor observations.

- Element type, classification, and species information
- All key attributes for that element type
- GPS coordinates and record date
- For visitor observations: photo, timestamp, user, and notes
- Link to observation review if the submission is pending

04

Sustainability staff

Observation Review

Opens when staff click a pending visitor observation. Staff verify, correct if needed, and mark it as reviewed.

- Full photo from the visitor submission
- Species name and element type — both editable by staff
- GPS coordinates — adjustable if inaccurate
- Notes from the visitor
- Verify — marks as verified and adds to the landscape inventory
- Flag — marks as needing follow-up without removing it
- Discard — removes the observation if it is invalid

5. Tech Stack

The platform is built on the following technologies. Each one was chosen to keep the system modern, maintainable, and consistent across the team.

Technology	What it does
Next.js	The main frontend framework for both the Field App and the Staff Dashboard
TypeScript	Keeps the codebase typed and easier to maintain across the whole team
Supabase	Handles the database, authentication, file storage, and real-time data
Leaflet.js + OSM	Renders the interactive arboretum map using real OpenStreetMap tiles
Docker	Ensures every team member runs the same development environment locally

6. In Scope vs. Out of Scope

This is the clearest way to align the team. If something is not in the left column, it does not get built in weeks 7–10.

✓ In Scope — Software Engineering	✗ Out of Scope — Later Modules
✓ Login with Supabase Auth	✗ User sign up and account creation
✓ Field App with 6 screens	✗ Native iOS and Android apps
✓ Real OSM map with Leaflet	✗ Offline map caching
✓ Observation pins from database	✗ Filtering and searching pins
✓ Tree detail bottom sheet	✗ Full species encyclopedia
✓ Photo capture via browser	✗ AI species identification (display only in MVP)
✓ Photo upload to Supabase Storage	✗ IoT sensor data integration
✓ Observation form with auto GPS	✗ QR code scanning on trees
✓ Staff dashboard with 4 screens	✗ ESG and sustainability report generation
✓ Landscape inventory as base layer	✗ Analytics: canopy coverage, carbon, pollinator maps
✓ Observation review and verify flow	✗ Historical landscape change detection
✓ Shared design system across both tools	✗ Deep learning or AI models
✓ Docker dev environment	✗ Multi-language support

The goal for Software Engineering

By the end of week 10, the full loop works end to end. A visitor logs in, submits a photo observation on the real arboretum map, and a staff member can open the dashboard, see the new submission, and verify it into the landscape inventory all on real devices, connected through Supabase.